CLINICAL TRIAL FAILURES · DISCONTINUATION RATE · NEUROLOGY PHASE 2,3 · STARTS 2015–2024

# BACE / γ-secretase: 3 molecules stopped early, all small molecules

**4 stopped trials, but 3 molecules — all small molecules: Lanabecestat, Atabecestat and Elenbecestat.** 3 of the stops were the trial's own verdict and 1 followed a decision taken elsewhere. Against 9.6% for all neurology trials with a resolved drug target and 6.5% across all neurology Phase 2,3 trials in the same window. The rate counts trials that stopped early for efficacy, safety or benefit–risk reasons. Trials that ran to completion and missed their primary endpoint are counted separately below, never in the rate.

**100.0%**
4 of 4 closed trials (95% CI 51.0%–100.0%)

**2 / 0 / 1**
efficacy / safety / both (adds to 4)

**100.0%**
of 4 trials have closed · 0 still open or unresolved

**50.0%**
stopped for a biological reason within 3 years of starting (95% CI 15.5%–94.2%; 2 still at risk; 3.5% area-wide)

— **Completed and missed the primary endpoint.** No trial in this segment has completed yet, so there is no posted result to read either way.

## How this compares

BACE / γ-secretase
100.0%
4/4 closed

all neurology trials with a resolved drug target
9.6%
65/676 closed

All neurology Phase 2,3
6.5%
101/1555 closed

**What is in this class:** 3 distinct experimental drugs across 4 trials (small molecule 4). Most tested: LANABECESTAT (2), ATABECESTAT (1), ELENBECESTAT (1). A class groups drugs by what they act on; check that the grouping is one you would make before reading the rate as a property of the mechanism.

**How much does this rest on one decision?** The 4 stops came from 3 sponsor-asset programmes across 3 sponsors; the largest (AstraZeneca) contributed 2; removing that programme's trials from both sides leaves 100.0%. Screened alongside 27 other segments in this area, the rate stays unusual after a 10% false-discovery correction (q=0.00299, Benjamini-Yekutieli). Read it as a screen worth checking against the underlying trials.

Bars show the share of closed trials stopped for biological reasons. Confidence intervals overlap where sample sizes are small — read the counts, not just the bars.

## By start cohort

| Trials started | Stops | Closed | Rate | 95% CI |
|---|---|---|---|---|
| Too few closed trials per cohort. | | | | |

## Where the stops sit

Efficacy / futility: 2, Efficacy and safety: 1, Benefit–risk: 1.
Sponsors with most stops: AstraZeneca (2), Eisai (1), Johnson & Johnson (1).

## The stopped trials

| Trial | Ph | Sponsor | Experimental drugs | Stopped | Type | Registry stop reason |
|---|---|---|---|---|---|---|
| NCT02956486 | 3 | Eisai | ELENBECESTAT | 2020-01 | Efficacy + safety | Due to an unfavorable risk-benefit ratio including no evidence of potential efficacy, and the adverse event profile of E2609 being worse than placebo. |
| NCT02569398 | 3 | Johnson & Johnson JNJ | ATABECESTAT | 2018-12 | Benefit–risk | Change in benefit-risk profile for individuals with early sporadic Alzheimer Disease because of elevations in liver enzymes in subjects receiving atabecestat |
| NCT02972658 | 3 | AstraZeneca AZN | LANABECESTAT | 2018-10 | Efficacy | As the feeder study (AZES) was stopped for futility after an independent assessment, this trial was also stopped. |

| Trial | Ph | Sponsor | Experimental drugs | Stopped | Type | Registry stop reason |
|---|---|---|---|---|---|---|
| NCT02783573 | 3 | AstraZeneca AZN | LANABECESTAT | 2018-09 | Efficacy | An independent assessment concluded the trial was not likely to meet the primary endpoint upon completion and therefore, trial stopped for futility. |

**Method**

Denominator: ClinicalTrials.gov interventional Phase 2,3 neurology trials started 2015–2024 that have closed (completed or terminated). Numerator: terminated trials whose registry stop reason is classified as biological (efficacy, safety or benefit–risk) by Classification V2 — held-out precision 95.5%, recall 95.3% (n=600). Drugs are linked to ChEMBL and the NCI Thesaurus; 45% of industry neurology trials in this window carry a resolved drug target, and a trial without one cannot enter a mechanism class. Intervals are Wilson 95%.

**Limits**

Not a failure rate: completed trials that missed their endpoints are not in the rate. They are counted separately where the sponsor posted a primary result that can be read — most did not, so that count is a floor — and programmes discontinued after a completed trial do not appear. Stop reasons are sponsor-reported. This is a closed-trial proportion, not a time-to-event analysis: only 100.0% of trials in this segment have closed, and a trial that stops early enters the denominator sooner than one that runs to completion, which can inflate the rate in immature segments. Recent cohorts have fewer closed trials, so their rates are less stable. Research signals, not clinical or investment advice.

**The whole cohort, trial by trial.** The diligence report names every trial behind these figures — the ones still running and the ones that completed and missed their primary endpoint included — and places your molecule against each one that failed. €99, delivered immediately: **clinicaltrialfailures.com/asset-check**

Sources: ClinicalTrials.gov (NLM); ChEMBL (EMBL-EBI, CC BY-SA 3.0); NCI Thesaurus (NCI); RxNorm/RxClass (NLM); SEC EDGAR. Classification, linkage and rates by Clinical Trial Failures. Rebuilt weekly; this brief covers trials started 2015–2024.